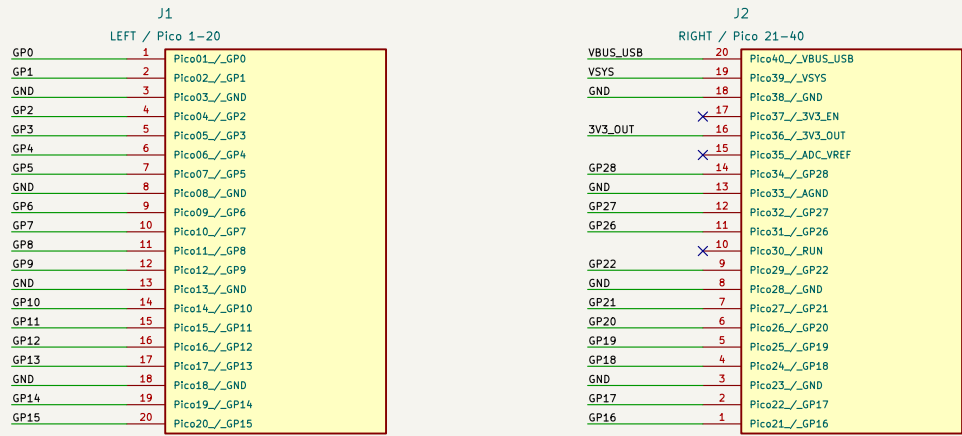


MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

REMOVABLE PICO: TOP VIEW, USB AT TOP

Two 1x20 female sockets: 2.54 mm pitch, 17.78 mm row spacing. Carrier contacts are passive.



J1 pin 1 = Pico 1 (USB end), J2 pin 1 = Pico 21 (opposite USB).

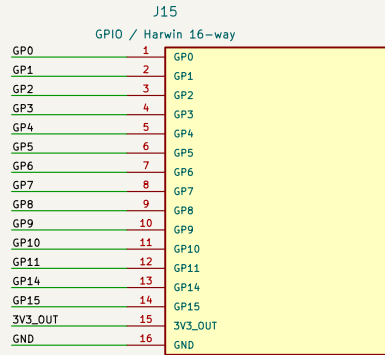
J2 pad 20 = Pico 40 at USB end: rotate J2 footprint 180 degrees relative to J1 in placement.

NC: Pico 30 RUN, 35 ADC_VREF, 37 3V3_EN, On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23-25/29 are internal, not expansion.

SSQ candidate: 1.02 mm drills / 1.52 mm pads. Verify finished holes and male-header fit.

FREE GPIO EXPANSION – CARRIER ORDER DEFINED



GP0/GP1: optional I2C SDA/SCL.

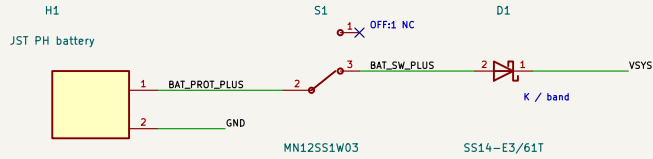
No I2C pull-ups fitted until bus/device choice.

REVIEW ONLY – NOT FOR FABRICATION

ERC checks carrier wiring only. Cell PCM, branch fault limits, module electronics and load/fit remain unverified.

Sources: Raspberry Pi Pico/Pico 2 datasheets, Figure 2 pin allocation; see ../reviews/schematic.md.

REV1 EXTERNAL-CHARGE BATTERY POWER



H1 JST PH: circuit 1=BAT_PROT_PLUS, circuit 2=GND.

Wired 1101 cassette approved; protected-cell fit pending.

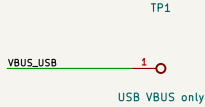
Branch fuse/current limit and PCM cutoff remain TBD.

S1 OFF Isolates battery; USB can still power Pico.

USB -> Pico onboard diode -> VSYS (inside Pico).

No carrier VBUS bridge; no onboard charger.

Remove cell for external charging; see power handoff.

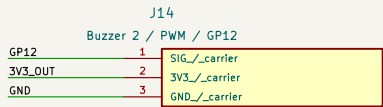
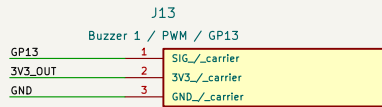
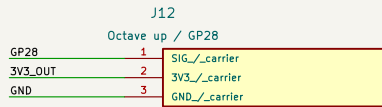
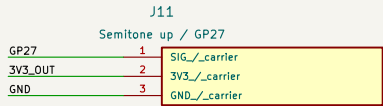
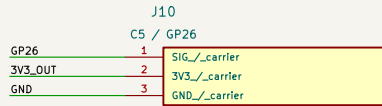
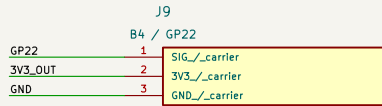
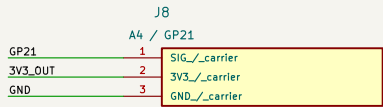
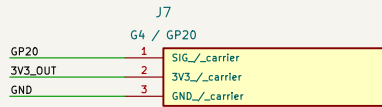
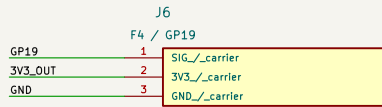
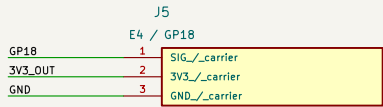
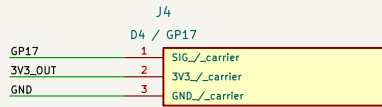
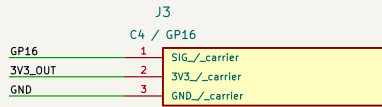


Cell/L1 charger are review targets, not approved parts.

TEN TOUCH / TWO BUZZER CARRIER CABLE HEADERS

J3-J14 carrier-defined order: 1=SIG, 2=3V3, 3=GND. Map cable to module by signal label.

Harwin carrier header footprints assigned. No direct module-mount/hole geometry; see cable contract.



SIGNAL / POWER LIMITS TO RESOLVE

Touch: TTP223 at 3.3 V, direct mode (TOG=0). Firmware may adapt to confirmed active-high/low.

ST0238 low-level driver: R1/R2 bias GP13/12 HIGH at reset; firmware HIGH idle still required.

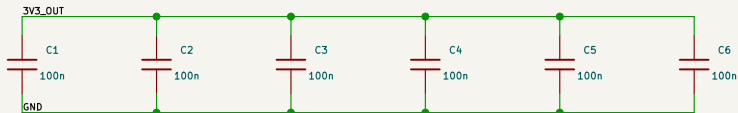
GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

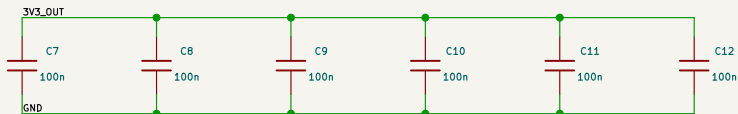
J1/J2: SSQ-120-01-G-5 candidate assigned; mating fit, USB/BOOTSEL and battery clearances TBD.

Reference may be active-low (AHLB=1). Confirm actual polarity/pin order before firmware changes.

C1-C12: 100nF / 50V X7R per carrier header



C1/J3 .. C12/J14: place locally; short GND return.



Review only – parts and power unresolved

Sheet: /
File: make_music.kicad_sch

Title: Make Music – provisional Pico carrier

Size: A3 Date: 2026-10-08

KiCad E.D.A. 10.0.6

Rev: P3

Id: 1/1